

PROJECT DOCUMENTATION

STEP BY STEP PROJECT DEVELOPMENT PROCEDURE

Project Title

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Step 1: Problem Identification

The housing market generates a huge amount of property data, including sale prices, house features, renovation history, basement area, bedrooms, bathrooms, and other details. Analyzing such large datasets manually is difficult and time-consuming.

Real estate professionals, investors, and home buyers require an easy way to understand market trends and identify factors affecting property prices. Without proper visualization, valuable insights remain hidden in raw data.

Therefore, this project focuses on building an interactive Tableau dashboard that transforms raw housing data into meaningful visualizations, helping users make informed real estate decisions.

Step 2: Problem Statement

To analyze residential housing market data using Tableau and identify the major factors influencing property sale prices through interactive dashboards, charts, and visual analytics.

Step 3: Objectives

The main objectives of this project are:

- Analyze housing market trends using Tableau.
 - Study the relationship between house features and sale price.
 - Evaluate the effect of house renovation on property value.
 - Compare structural features such as bedrooms, bathrooms, floors, and basement area.
 - Build an interactive dashboard for better data exploration.
 - Present insights through Tableau Story.
 - Publish the dashboard on Tableau Public.
 - Integrate the dashboard into a Flask web application.
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Step 4: Data Collection

The dataset used for this project was collected in CSV format and contains detailed information about residential properties.

Dataset Details

- **Dataset Name:** Transformed_Housing_Data2.csv
- **Number of Records:** 21,609
- **Number of Columns:** 31
- **File Type:** CSV
- **Missing Values:** None

Important Attributes

- Sale Price
- Bedrooms
- Bathrooms
- Floors
- House Age
- Basement Area
- Living Area
- Lot Area
- Waterfront
- View
- Condition
- Grade
- Year Built
- Year Renovated
- Years Since Renovation

The dataset provides sufficient information to analyze the relationship between housing characteristics and sale prices.

Step 5: Data Cleaning and Preparation

Before creating visualizations, the dataset was cleaned and prepared to improve data quality.

Data Cleaning Process

- Imported the CSV dataset into Tableau.
- Verified that all 31 columns had the correct data types.
- Confirmed that the dataset contained no missing values.
- Checked for duplicate records.
- Formatted numerical fields such as Sale Price.

- Created Sale Price bins for better visualization.
- Organized fields into Dimensions and Measures.
- Verified calculated fields such as House Age and Years Since Renovation.

The dataset was then ready for visualization.

Step 6: Data Visualization

Several worksheets were created to analyze different aspects of the housing market.

Worksheet 1 – Housing Market Overview

A single KPI worksheet was created to display three important housing market indicators:

- Total Number of Houses
- Average Sale Price
- Total Basement Area

These KPIs provide a quick summary of the dataset and help users understand the overall housing market before exploring detailed visualizations.

Purpose:

- Display the total number of residential properties.
- Show the average selling price of houses.
- Present the total basement area across all properties.
- Provide an overview of key housing statistics in a single view.

Worksheet 2 – Total Sales by Years Since Renovation

Analyzes how property sale prices vary according to renovation history.

Purpose:

- Evaluate the impact of renovations on sale prices.
- Compare renovated and non-renovated properties.

Worksheet 3 – Distribution of House Age by Renovation Status

Shows the proportion of renovated and non-renovated houses across different house ages.

Purpose:

- Understand renovation patterns.

- Analyze the relationship between house age and renovation status.
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Worksheet 4 – House Age Distribution by Bathrooms, Bedrooms, and Floors

Compares house age with important structural features such as bedrooms, bathrooms, and floors.

Purpose:

- Analyze housing characteristics.
 - Compare structural features across different age groups.
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Step 7: Dashboard Development

After creating individual worksheets, all visualizations were combined into a single interactive dashboard.

Dashboard Features

- KPI Cards
- Bar Charts
- Pie Charts
- Interactive Filters
- Legends
- Dynamic Tooltips
- Responsive Layout

The dashboard allows users to interactively explore housing data and compare different property features.

Step 8: Story Creation

A Tableau Story was created to present the project findings in a logical sequence.

Story Points

1. Total Houses
2. Average Sale Price
3. Basement Area Analysis
4. Sales by Years Since Renovation

5. House Age Distribution
6. House Features Analysis
7. Final Housing Market Insights

The story guides users from basic information to advanced market insights.

Step 9: Dashboard Testing

The dashboard was tested to ensure that it functions correctly.

Testing Activities

- Verified successful dataset loading.
 - Checked dashboard responsiveness.
 - Tested interactive filters.
 - Validated KPI values.
 - Verified chart accuracy.
 - Tested tooltip functionality.
 - Ensured smooth navigation between story pages.
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Step 10: Publishing

The completed Tableau workbook was published on Tableau Public.

Publishing Steps

- Signed in to Tableau Public.
 - Published the workbook.
 - Generated the public dashboard URL.
 - Verified that the dashboard was accessible online.
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Step 11: Flask Web Integration

The published Tableau dashboard was embedded into a Flask web application using the Tableau Embed API.

Features

- Web-based dashboard access.

- Interactive visualizations.
 - Responsive interface.
 - Easy navigation.
 - Users can access dashboards without opening Tableau Desktop.
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Step 12: Results

The project successfully identified several housing market trends.

Key Findings

- The dataset contains over **21,000 residential properties**.
 - Renovated houses generally have higher sale prices than non-renovated houses.
 - House age influences structural characteristics.
 - Basement area contributes to property analysis.
 - Interactive dashboards simplify the understanding of complex housing data.
 - Tableau enables faster and more effective decision-making through visual analytics.
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Step 13: Conclusion

This project demonstrates how Tableau transforms raw housing data into meaningful visual insights. The interactive dashboards and storytelling features help users analyze housing prices, compare property characteristics, and understand market trends efficiently. The project supports informed real estate decision-making through data visualization.

Step 14: Future Scope

Future enhancements to this project include:

- Integrating real-time housing market data.
- Predicting house prices using Machine Learning algorithms.
- Adding interactive geographic maps.
- Comparing neighborhoods and locations.
- Deploying the application on cloud platforms.
- Developing AI-based recommendations for buyers and sellers.
- Creating a mobile-friendly dashboard for easier access.